1.10 Pre-Quiz: Statistics — Markscheme

1. [Maximum mark: 8]

   Mid-interval values $1, 3, 5, 7, 9$ with frequencies $6, 14, 20, 12, 8$; $n = 60$.

   GDC: *enter the mid-interval values in one list and the frequencies in a second list, then run* `One-Variable Statistics` *with the frequency list.*

   (a) Continuous **A1**

   Time is measured, not counted. Accept a reason such as "time can take any value in an interval"; no reason is required.

   (b) $4 \le h < 6$ **A1**

   *The class, not its frequency: "*20*" alone earns A0. Accept "*4 *to* 6 *hours". The mid-interval value* 5 *alone earns A0.*

   (c) $\bar{x} = \dfrac{6(1) + 14(3) + 20(5) + 12(7) + 8(9)}{60}$ **(M1)**

   $= \dfrac{304}{60} = 5.0666\ldots = 5.07$ hours (3 s.f.) **A1**

   M1 for using the mid-interval values with the frequencies, shown as a sum or as the GDC lists. Accept 5.067 *or* $\frac{76}{15}$*. Using the lower class boundaries (*$0, 2, 4, 6, 8$*, giving* 4.07*) or upper boundaries (giving* 6.07*) earns M0A0. The mean of the frequencies,* 12*, earns no marks.*

   (d) The exact number of hours for each student is not known; each student is taken to be at the mid-interval value of their class. **R1**

   R1 for either idea: the raw data are not given (only the classes), or mid-interval values were used in place of the actual values.

   (e) $\sigma = 2.3371\ldots = 2.34$ hours (3 s.f.) **A1**

   Do not accept $s_x = 2.3568\ldots$ *(*2.36*), the sample standard deviation.*

   (f) $\dfrac{112}{480} \times 60$ **M1**

   $= 14$ students **A1**

   M1 for the Grade 12 proportion of the school applied to 60*, however written (e.g.* $\frac{60}{480} = \frac{1}{8}$*, then* $\frac{112}{8}$*).*

2. [Maximum mark: 9]

Ordered: $21, 22, 23, 24, 25, 25, 26, \mathbf{26}, 27, 27, 28, 29, 30, 31, 44 \quad (n = 15)$.

GDC: *the TI-Nspire excludes the median from each half, so Q_1 is the median of the lowest seven values and Q_3 the median of the highest seven.*

(a) Median $= 26$ **A1**

$Q_1 = 24, \quad Q_3 = 29$ **A1**

Both quartiles are needed for the second A1. Accept $Q_1 = 24.5$ and $Q_3 = 28.5$ from a hand method that includes the median in each half; feedback: the GDC convention is the one to use.

(b) IQR $= 29 - 24 = 5$ **A1**

FT: award A1 for $Q_3 - Q_1$ from the candidate's own quartiles. The range, $44 - 21 = 23$, earns no mark.

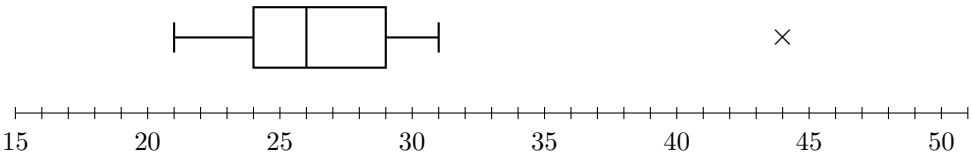
(c) $Q_3 + 1.5 \times \text{IQR} = 29 + 1.5 \times 5$ **M1**

$= 36.5$ **A1**

$44 > 36.5$, so 44 is an outlier. **R1**

M1 for $1.5\times$ their IQR added to their Q_3. R1 for the comparison with 44 stated, not just the fence value. Adding $1.5 \times$ IQR to the median or the mean earns no marks. The lower fence, $24 - 7.5 = 16.5$, is not required. FT: award all three marks from the candidate's own Q_3 and IQR, provided their fence is below 44.

(d)



Box from $Q_1 = 24$ to $Q_3 = 29$ with the median line at 26 **A1**

Whiskers to 21 and 31 **A1**

Outlier 44 marked with a cross **A1**

Award each mark independently; plotted values within half a small division (± 0.5 minute) are accepted. A whisker drawn to 44 or to 36.5 loses the second A1. FT: award the first A1 for a box drawn at the candidate's own quartiles from (a).

3. [Maximum mark: 9]

80 students; the curve passes through $(20, 8)$, $(30, 20)$, $(40, 40)$, $(55, 60)$ and $(60, 65)$.

Readings within one small square of the values below are accepted (± 2.5 *seconds on the time axis,* ± 1 *student on the cumulative frequency axis), and any value following from the candidate's own readings earns the A1. Feedback: draw the reading lines on the graph.*

(a) Median: cumulative frequency 40 gives $t = 40$ seconds **A1**

(b) $Q_1 = 30$ (at 20), $\quad Q_3 = 55$ (at 60) **(M1)**

$\text{IQR} = 55 - 30 = 25$ seconds **A1**

M1 for reading the times at cumulative frequencies 20 *and* 60 *(or for either quartile correct). Reading at* 25 *and* 75 *(percent of* 100*) earns M0.*

(c) $80 - 65$ **(M1)**

$= 15$ students **A1**

M1 for reading the cumulative frequency at $t = 60$ *and subtracting from* 80*. The reading* 65 *alone, as a final answer, earns M1A0.*

(d) 10% of $80 = 8$ **(M1)**

Cumulative frequency 8 gives $t = 20$ seconds **A1**

M1 for 8 *(or for reading the time at cumulative frequency* 8*). Reading at cumulative frequency* 72 *(the slowest* 10%*,* $t = 70$*) earns M1A0.*

(e) $p = 40 - 8 = 32$ **A1**

$q = 65 - 40 = 25$ **A1**

The two marks are independent. Accept $q = 80 - 8 - 32 - 15 = 25$*. FT: award the A1 for q from the candidate's own p when q is found by subtraction from* 80*.*

4. [Maximum mark: 9]

$n = 8, \quad \bar{x} = 7, \quad \bar{y} = 68.$

GDC: *enter x and y in two lists and run* `Linear Regression (mx+b)`.

(a) $r = 0.98815\ldots = 0.988$ (3 s.f.) **A1**

Do not accept $r^2 = 0.976$, which sits beside r on the regression screen.

(b) Strong, positive (linear) correlation **A1A1**

A1 for "strong" (accept "very strong"), A1 for "positive"; the two marks are independent. "Moderate" or "weak" earns A0 for the strength. FT from the candidate's r in (a).

(c) $y = 3.72x + 42.0$ **A1A1**

A1 for $a = 3.72$, A1 for $b = 42.0$; the two marks are independent. Accept $a = 3.7173\ldots$, $b = 41.978\ldots$ (or 42), or either value to more figures. The reversed line $x = 0.263y - 10.9$ earns A0A0.

(d) For each extra hour of revision, the test score increases by about 3.72 points. **A1**

The answer must give the rate (about 3.7 points) per hour, in context. "Students who revise more score higher" restates the correlation without the rate and earns A0. FT from the candidate's a in (c).

(e) $y = 3.72(10) + 42.0$ **(M1)**

$= 79.152\ldots \approx 79.2$ **A1**

Accept any value consistent with the candidate's own correctly-rounded coefficients, including 79.152… (full precision) and 79.2 (from 3.72 and 42.0), and either rounded to 79. FT from the candidate's line in (c). ***Feedback:*** *store the coefficients in the GDC (or use the regression function* `f1(10)`*) and round only the final answer.*

(f) 20 hours is outside the range of the data (2 to 12 hours), so the estimate would be an extrapolation and is not reliable. **R1**

R1 for stating that 20 lies outside the data range (or "above the greatest number of hours recorded"). Accept the observation that the line gives 116, which is impossible for a score out of 100, as a supporting reason. "Revising too much might not help" with no reference to the data earns R0.